

Project phase I submission guide

Goal

For Phase I, your objectives are:

- Implement a Cardinal Router
- Connect the routers to form a 4×4 mesh network
- Implement a cardinal NIC

Synthesis is not required in this phase, but you must follow synthesizable Verilog coding style, as it will be required in later phases of the project.

Things to submit

- Cardinal router and 4×4 Mesh
 - gold_router.v — top-level router design
 - All Verilog design files needed to build the router
 - (Optional) The router-level testbench tb_gold_router.v
 - The 4×4 mesh interconnect file gold_mesh.v
 - The top-level testbench tb_gold_mesh.v
 - Simulation output logs, as specified in Lecture 9's PDF
- Cardinal NIC
 - All Verilog design and testbench files used for your NIC implementation

Note: Please ensure that all module port names strictly follow the input/output naming conventions shown in the specification (see attached diagrams).

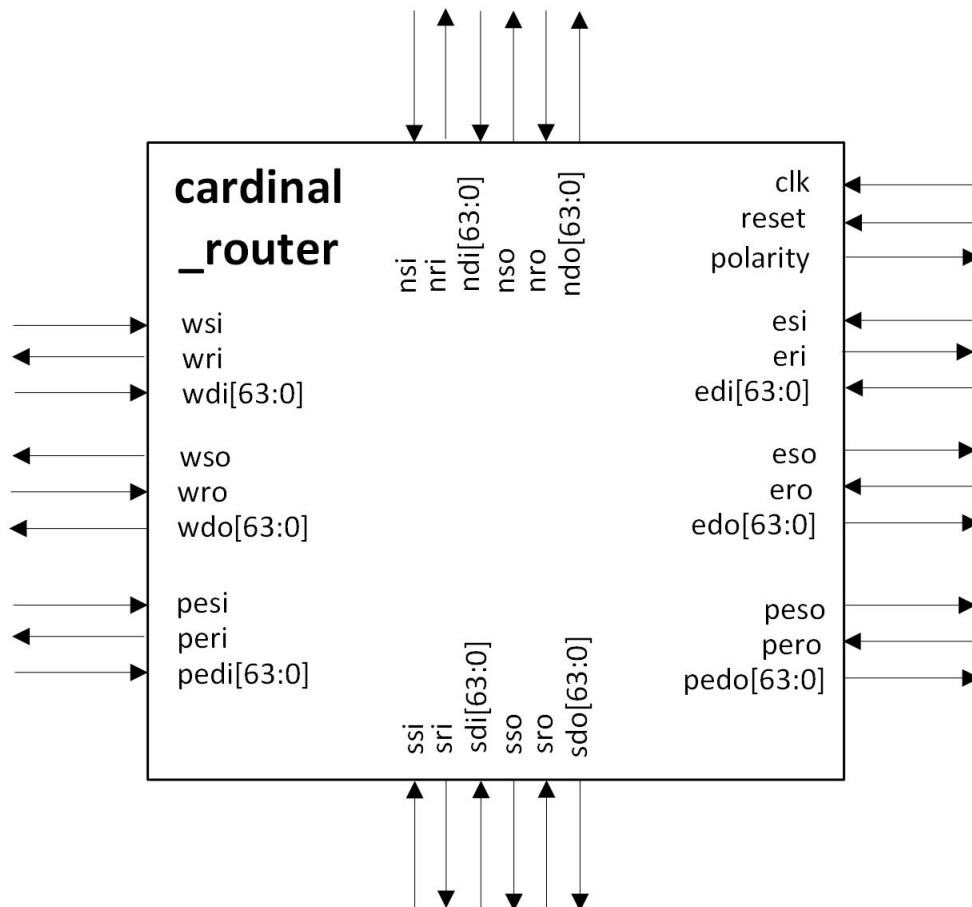


Figure 1: Cardinal Router External Interface

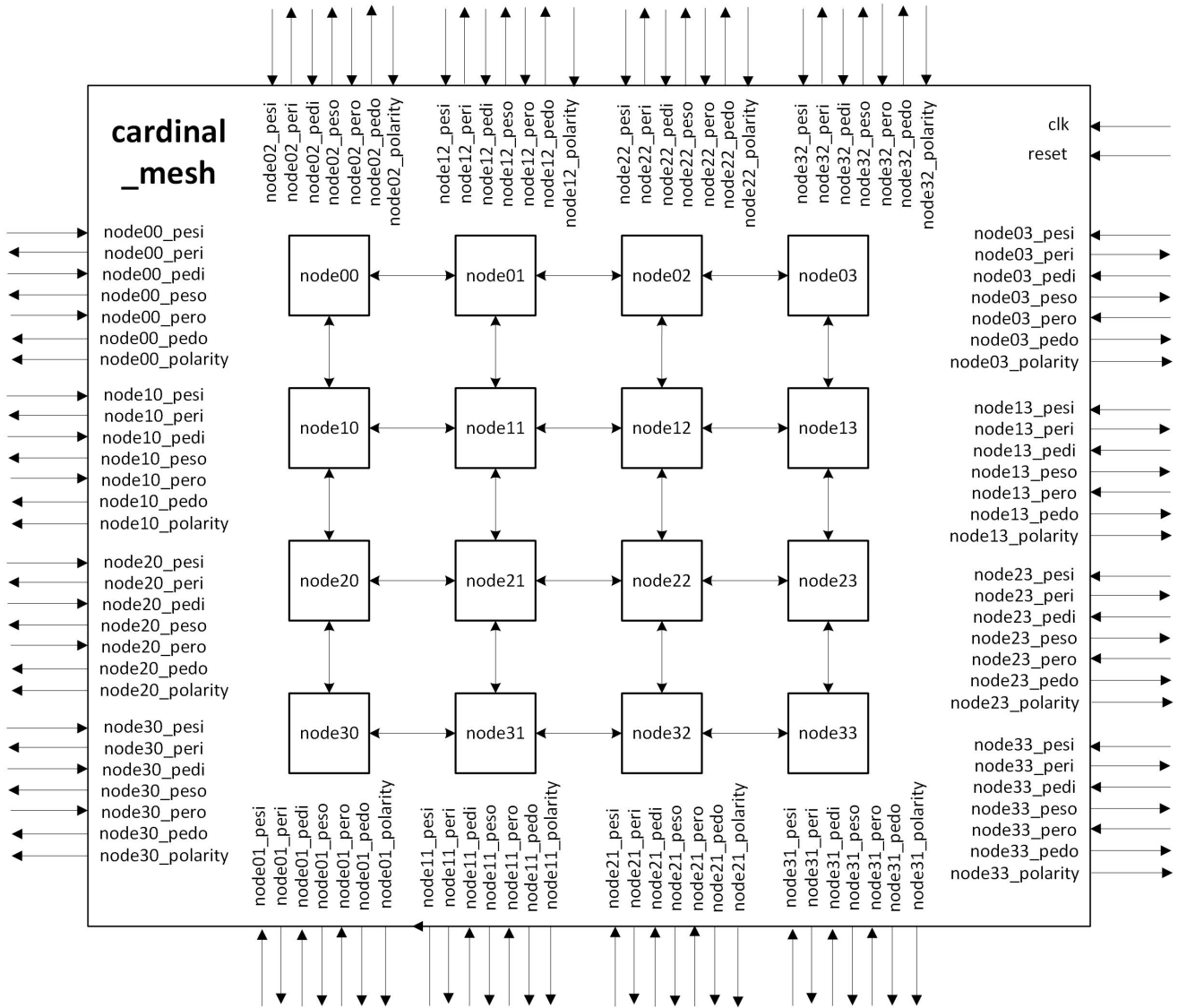


Figure 2: Cardinal Mesh Interface

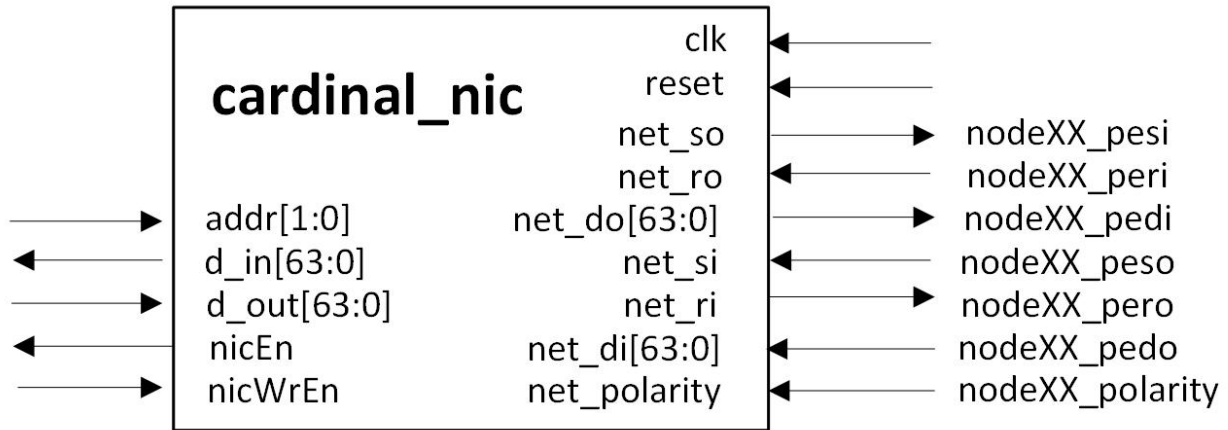


Figure 3: Cardinal NIC Interface

Supplementary material (Optional but Recommended)

Although module-level router verification is not required, it is strongly recommended. Partial credit will be awarded for showing correct module-level functionality, and it's good practice to validate your design at multiple hierarchy levels.

For this part, you may include but not limited to:

- Individual module-level testbenches
- A short report including results of your simulation (waveform or .txt), each with a short explanation. Your results/test cases should at least cover:
 - Cardinal mesh router
 - One waveform showing reset signal operation
 - Correct hop value updates
 - At least six test cases covering different routing and switching conditions
 - Waveforms showing router-to-router handshaking, both with and without blocking (Refer to Figures 4–5 in the project PDF)
 - At least one test case showing arbiter behavior when contention occurs, clearly showing priority rotation
 - Cardinal NIC
 - One waveform showing the correct operation of the reset signal
 - Waveform showing the handshake between NIC and router, both with and without blocking. (You may refer to Figure 3 in the instruction)
 - Waveform showing interactions between processor and NIC for both load and store operations each with buffer being both available and unavailable. (Total 4 cases)

Submission format

Please submit a single ZIP file containing all parts.

Name your folder as: Phase_<PhaseNumber>_Group_<GroupNumber>

Example: Phase_1_Group_1.zip

Only one submission per team is required.

- One for your Cardinal router SimVision folder, you may include your log file inside this folder as well. (Directly under the root folder)
- One for your Cardinal NIC SimVision folder
- Your report with explanation to your results

General guidelines

- Start early and work together with your teammate.
- Version control and synchronization platform such as GitHub is recommended.
- Add comment to your code when needed, write clean and formatted code. This will not only help us on grading but will also help your teammate to better understand your work.
- Write modularized RTL — how you structure submodules is up to you, but keep hierarchy clear.